

Assignment No 3

Smart Grievance Handling System

1) Single - Row Subquery

```
mysql> SELECT title, priority_score
-> FROM Issue
-> WHERE priority_score = (
->     SELECT MAX(priority_score)
->     FROM Issue
-> );
```

title	priority_score
Pothole near highway	90
Water leakage	90
Drainage blockage	90
Tree fallen	90

4 rows in set (0.00 sec)

```
mysql> SELECT title, priority_score
-> FROM Issue
-> WHERE priority_score > (
->     SELECT AVG(priority_score)
->     FROM Issue
-> );
```

title	priority_score
Pothole near highway	90
Water leakage	90
Drainage blockage	90
Tree fallen	90

4 rows in set (0.00 sec)

```
mysql> SELECT title, priority_score
-> FROM Issue
-> WHERE priority_score < (
->     SELECT AVG(priority_score)
->     FROM Issue
-> );
```

title	priority_score
Street light not working	30
Illegal parking	30
Garbage not collected	60
Road damage	60
Public toilet unclean	60

5 rows in set (0.00 sec)

```
mysql> SELECT title, priority_score
-> FROM Issue
-> WHERE priority_score >= (
->     SELECT AVG(priority_score)
->     FROM Issue
[ -> );
```

title	priority_score
Pothole near highway	90
Water leakage	90
Drainage blockage	90
Tree fallen	90

4 rows in set (0.00 sec)

```
mysql> SELECT title, priority_score
-> FROM Issue
-> WHERE priority_score <= (
->     SELECT AVG(priority_score)
->     FROM Issue
[ -> );
```

title	priority_score
Street light not working	30
Illegal parking	30
Garbage not collected	60
Road damage	60
Public toilet unclean	60

5 rows in set (0.00 sec)

```
mysql> SELECT title, priority_score
-> FROM Issue
-> WHERE priority_score <> (
->     SELECT MIN(priority_score)
->     FROM Issue
[ -> );
```

title	priority_score
Pothole near highway	90
Garbage not collected	60
Water leakage	90
Road damage	60
Drainage blockage	90
Tree fallen	90
Public toilet unclean	60

7 rows in set (0.00 sec)

2) Multiple - Row Subquery

```
mysql> SELECT title
-> FROM Issue
-> WHERE citizen_id IN (
->     SELECT citizen_id
->     FROM Citizen
->     WHERE area = 'Akurdi'
[ -> );
```

```
+-----+
| title          |
+-----+
| Pothole near highway |
| Road damage      |
| Public toilet unclean |
+-----+
3 rows in set (0.00 sec)
```

```
mysql> SELECT title
-> FROM Issue
-> WHERE citizen_id NOT IN (
->     SELECT citizen_id
->     FROM Citizen
->     WHERE area = 'Akurdi'
[ -> );
```

```
+-----+
| title          |
+-----+
| Garbage not collected |
| Water leakage        |
| Street light not working |
| Drainage blockage     |
| Illegal parking      |
| Tree fallen          |
+-----+
6 rows in set (0.00 sec)
```

```
mysql> SELECT title, priority_score
-> FROM Issue
-> WHERE priority_score > ANY (
->     SELECT priority_score
->     FROM Issue
->     WHERE severity = 'Low'
[ -> );
```

title	priority_score
Pothole near highway	90
Garbage not collected	60
Water leakage	90
Road damage	60
Drainage blockage	90
Tree fallen	90
Public toilet unclean	60

7 rows in set (0.00 sec)

```
mysql> SELECT title, priority_score
-> FROM Issue
-> WHERE priority_score > SOME (
->     SELECT priority_score
->     FROM Issue
->     WHERE severity = 'Low'
[ -> );
```

title	priority_score
Pothole near highway	90
Garbage not collected	60
Water leakage	90
Road damage	60
Drainage blockage	90
Tree fallen	90
Public toilet unclean	60

7 rows in set (0.00 sec)

```
mysql> SELECT title, priority_score
-> FROM Issue
-> WHERE priority_score > ALL (
->     SELECT priority_score
->     FROM Issue
->     WHERE severity = 'Low'
[ -> );
```

title	priority_score
Pothole near highway	90
Garbage not collected	60
Water leakage	90
Road damage	60
Drainage blockage	90
Tree fallen	90
Public toilet unclean	60

7 rows in set (0.00 sec)

3) Exists Operator

```
mysql> SELECT name
-> FROM Citizen C
-> WHERE EXISTS (
->     SELECT 1
->     FROM Issue I
->     WHERE I.citizen_id = C.citizen_id
[   -> );
+-----+
| name |
+-----+
| Vijay |
| Sahil |
| Atharva |
| Omraje |
| Prachi |
| Sanskar |
| Aniket |
| Pratima |
| Srushti |
+-----+
9 rows in set (0.00 sec)

mysql> SELECT name
-> FROM Citizen C
-> WHERE NOT EXISTS (
->     SELECT 1
->     FROM Issue I
->     WHERE I.citizen_id = C.citizen_id
[   -> );
Empty set (0.00 sec)
```

4) Multiple Column SubQuery

```
mysql> SELECT title, area
-> FROM Issue
-> WHERE (area, severity) IN (
->     SELECT area, severity
->     FROM Issue
[   -> );
+-----+-----+
| title | area |
+-----+-----+
| Pothole near highway | Akurdi |
| Garbage not collected | Nigdi |
| Water leakage | Pimpri |
| Street light not working | Chinchwad |
| Road damage | Akurdi |
| Drainage blockage | Nigdi |
| Illegal parking | Pimpri |
| Tree fallen | Chinchwad |
| Public toilet unclean | Akurdi |
+-----+-----+
9 rows in set (0.01 sec)
```

5) Correlated SubQuery

```
mysql> SELECT title, priority_score, area
-> FROM Issue I1
-> WHERE priority_score > (
->     SELECT AVG(priority_score)
->     FROM Issue I2
->     WHERE I2.area = I1.area
-> );
```

title	priority_score	area
Pothole near highway	90	Akurdi
Water leakage	90	Pimpri
Drainage blockage	90	Nigdi
Tree fallen	90	Chinchwad

4 rows in set (0.00 sec)

6) Nested SubQuery

```
mysql> SELECT title
-> FROM Issue
-> WHERE citizen_id = (
->     SELECT citizen_id
->     FROM Citizen
->     WHERE area = 'Pimpri'
->     LIMIT 1
-> );
```

title
Water leakage

1 row in set (0.00 sec)

7) Joins

i) Inner Join

```
mysql> SELECT I.issue_id,  
-> I.title,  
-> I.area,  
-> I.severity,  
-> I.priority_score,  
-> I.status,  
-> C.name AS citizen_name  
-> FROM Issue I  
-> INNER JOIN Citizen C  
-> ON I.citizen_id = C.citizen_id;
```

issue_id	title	area	severity	priority_score	status	citizen_name
1	Pothole near highway	Akurdi	High	90	Open	Vijay
2	Garbage not collected	Nigdi	Medium	60	Open	Sahil
3	Water leakage	Pimpri	High	90	Open	Atharva
4	Street light not working	Chinchwad	Low	30	Open	Omraje
5	Road damage	Akurdi	Medium	60	Open	Prachi
6	Drainage blockage	Nigdi	High	90	Open	Sanskar
7	Illegal parking	Pimpri	Low	30	Open	Aniket
8	Tree fallen	Chinchwad	High	90	Open	Pratima
9	Public toilet unclean	Akurdi	Medium	60	Open	Srushti

9 rows in set (0.00 sec)

ii) Left Join

```
mysql> SELECT I.issue_id,  
-> I.title,  
-> C.name  
-> FROM Issue I  
-> LEFT JOIN Citizen C  
-> ON I.citizen_id = C.citizen_id;
```

issue_id	title	name
1	Pothole near highway	Vijay
2	Garbage not collected	Sahil
3	Water leakage	Atharva
4	Street light not working	Omraje
5	Road damage	Prachi
6	Drainage blockage	Sanskar
7	Illegal parking	Aniket
8	Tree fallen	Pratima
9	Public toilet unclean	Srushti
10	Test Issue	NULL

10 rows in set (0.01 sec)

iii) Right Join

```
mysql> SELECT I.issue_id,
->      I.title,
->      C.name
-> FROM Issue I
-> RIGHT JOIN Citizen C
-> ON I.citizen_id = C.citizen_id;
```

issue_id	title	name
1	Pothole near highway	Vijay
2	Garbage not collected	Sahil
3	Water leakage	Atharva
4	Street light not working	Omraje
5	Road damage	Prachi
6	Drainage blockage	Sanskar
7	Illegal parking	Aniket
8	Tree fallen	Pratima
9	Public toilet unclean	Srushti
NULL	NULL	ExtraUser
NULL	NULL	Akash

11 rows in set (0.00 sec)

iv) Full Join

```
mysql> SELECT I.issue_id,
->      I.title,
->      C.name
-> FROM Issue I
-> LEFT JOIN Citizen C
-> ON I.citizen_id = C.citizen_id
->
-> UNION
->
-> SELECT I.issue_id,
->      I.title,
->      C.name
-> FROM Issue I
-> RIGHT JOIN Citizen C
-> ON I.citizen_id = C.citizen_id;
```

issue_id	title	name
1	Pothole near highway	Vijay
2	Garbage not collected	Sahil
3	Water leakage	Atharva
4	Street light not working	Omraje
5	Road damage	Prachi
6	Drainage blockage	Sanskar
7	Illegal parking	Aniket
8	Tree fallen	Pratima
9	Public toilet unclean	Srushti
10	Test Issue	NULL
NULL	NULL	ExtraUser
NULL	NULL	Akash

12 rows in set (0.00 sec)

v) Cross Join

```
mysql> SELECT C.name,  
->         I.title  
-> FROM Citizen C  
-> CROSS JOIN Issue I  
-> LIMIT 18;  
+-----+-----+  
| name | title |  
+-----+-----+  
| Vijay | Test Issue |  
| Vijay | Public toilet unclean |  
| Vijay | Tree fallen |  
| Vijay | Illegal parking |  
| Vijay | Drainage blockage |  
| Vijay | Road damage |  
| Vijay | Street light not working |  
| Vijay | Water leakage |  
| Vijay | Garbage not collected |  
| Vijay | Pothole near highway |  
| Sahil | Test Issue |  
| Sahil | Public toilet unclean |  
| Sahil | Tree fallen |  
| Sahil | Illegal parking |  
| Sahil | Drainage blockage |  
| Sahil | Road damage |  
| Sahil | Street light not working |  
| Sahil | Water leakage |  
+-----+-----+  
18 rows in set (0.00 sec)
```

vi) Self Join

```
mysql> SELECT I1.title AS Issue1,  
->      I2.title AS Issue2,  
->      I1.area  
-> FROM Issue I1  
-> JOIN Issue I2  
-> ON I1.area = I2.area  
[ -> AND I1.issue_id <> I2.issue_id;
```

Issue1	Issue2	area
Public toilet unclean	Pothole near highway	Akurdi
Road damage	Pothole near highway	Akurdi
Drainage blockage	Garbage not collected	Nigdi
Illegal parking	Water leakage	Pimpri
Tree fallen	Street light not working	Chinchwad
Public toilet unclean	Road damage	Akurdi
Pothole near highway	Road damage	Akurdi
Garbage not collected	Drainage blockage	Nigdi
Water leakage	Illegal parking	Pimpri
Street light not working	Tree fallen	Chinchwad
Road damage	Public toilet unclean	Akurdi
Pothole near highway	Public toilet unclean	Akurdi

12 rows in set (0.01 sec)

vii) Join with WHERE condition

```
mysql> SELECT C.name,  
->      I.title,  
->      I.area  
-> FROM Citizen C  
-> INNER JOIN Issue I  
-> ON C.citizen_id = I.citizen_id  
[ -> WHERE I.area = 'Akurdi';
```

name	title	area
Vijay	Pothole near highway	Akurdi
Prachi	Road damage	Akurdi
Srushti	Public toilet unclean	Akurdi

3 rows in set (0.01 sec)

viii) Join with GROUP BY

```
mysql> SELECT C.name,  
->          COUNT(I.issue_id) AS total_issues  
-> FROM Citizen C  
-> LEFT JOIN Issue I  
-> ON C.citizen_id = I.citizen_id  
[ -> GROUP BY C.name;
```

name	total_issues
Vijay	2
Sahil	2
Atharva	1
Omraje	1
Prachi	1
Sanskar	1
Aniket	1
Pratima	1
Srushti	1
ExtraUser	0
Akash	0

11 rows in set (0.01 sec)

ix) Join with HAVING

```
mysql> SELECT C.name,  
->          COUNT(I.issue_id) AS total_issues  
-> FROM Citizen C  
-> INNER JOIN Issue I  
-> ON C.citizen_id = I.citizen_id  
-> GROUP BY C.name  
[ -> HAVING COUNT(I.issue_id) > 1;
```

name	total_issues
Vijay	2
Sahil	2

2 rows in set (0.00 sec)